1 Motivation and Objectives

Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, highlighting the critical need for effective continuous monitoring and early detection. Recent advances in wearable technologies have made electrocardiogram (ECG) monitoring increasingly accessible in daily life.

However, most existing wearable ECG systems are limited to short-duration recordings, typically around one minute, and provide only basic real-time analysis, such as heart rate estimation. While some devices support arrhythmia detection, their functionality is often primarily limited to detecting persistent atrial fibrillation (AF), without the capability to identify more complex or transient conditions.

Atrial fibrillation (AF) is commonly classified into paroxysmal, persistent, and permanent forms according to episode duration and termination behavior [1]. Paroxysmal AF refers to episodes that terminate spontaneously, typically within several days, whereas persistent and permanent AF require clinical intervention or cannot be terminated. Due to its intermittent nature, paroxysmal AF is particularly challenging to detect using conventional short-duration monitoring systems, and such episodes are therefore frequently missed. This limitation is critical, as early detection of transient arrhythmias and comprehensive cardiovascular risk assessment are essential for timely medical intervention and long-term disease prevention.

Therefore, we propose an integrated framework that enables long-term ECG monitoring and incorporates two components: an atrial fibrillation detection module and a cardiovascular disease prediction model. The AF detection module focuses on identifying both paroxysmal and persistent atrial fibrillation from continuous ECG signals, while the cardiovascular disease prediction model provides a comprehensive assessment of cardiovascular conditions based on extracted features. Our proposed system offers a more complete solution for both real-time arrhythmia detection and cardiovascular risk evaluation.

2 System Design

2.1 Key Innovations

The proposed system introduces a unified framework for continuous ECG monitoring and analysis, integrating long-term data acquisition, real-time signal processing, and edge-based inference. Its key features are summarized as follows:

- **Interpretable physiological indicators**: The system incorporates cardiovascular risk assessment derived from ECG features, transforming raw signals into meaningful and interpretable outputs accessible to non-specialist users. In addition, it supports the export of ECG signal plots to facilitate subsequent review by healthcare professionals.
- **Long-term continuous monitoring**: Supports sustained ECG acquisition beyond the short-duration measurements of typical wearable devices.
- **Comprehensive AF detection**: Capable of detecting both paroxysmal and persistent AF.
- **Multi-condition ECG acquisition**: Enables ECG monitoring under both resting and exercise conditions, supporting more realistic usage scenarios.

The system is implemented using an edge-based architecture to achieve real-time inference with low latency and enhanced data privacy.

2.2 Comparison with Existing Solutions

Our system enables long-term, continuous ECG monitoring, representing a substantial advancement over typical commercial wearable devices that are generally restricted to short-duration recordings.

In addition, the system incorporates cardiovascular risk assessment, thereby enhancing the interpretability and clinical relevance of the acquired ECG signals.

Regarding atrial fibrillation (AF) detection, the proposed system is capable of identifying both paroxysmal and persistent AF, whereas many commercial devices primarily focus on heart rate monitoring or are limited to detecting persistent AF.

Furthermore, the system is specifically designed for wearable, continuous operation and supports ECG acquisition under both resting and exercise conditions. In contrast, many existing commercial devices rely on finger-contact measurements and are therefore limited to short-term, resting-state recordings.

However, our proposed system still has its limitations: our system still only supports single-lead ECG and might not be representitive enough for cardiovascular disease (We've planned to expand to multi-lead ECG in the future to increase the amount of information for risk assessment). Also, our system still relies on gel-electrode contact for ECG measurement.

Comparison Metrics	Proposed System	Existing Wearable Devices
ECG Monitoring Duration	Long-term continuous monitoring	Short-duration recordings

ECG Acquisition Conditions	Resting and exercise ECG	Primarily resting ECG
Measurement Method	Gel electrodes	Dry electrodes (finger-contact required)
Analytical Functionality	ECG feature analysis, risk assessment, and AF detection	Heart rate monitoring or limited AF detection
AF Detection Capability	Paroxysmal and persistent AF	Persistent AF detection
Computational Architecture	Edge-based real-time inference with enhanced privacy	Device-dependent; often limited on-device processing

Table 1: Functional comparison with commercial products

2.2.1 Clinical ECG systems

2.2.2 Smartwatch ECG

2.3 Potential Applications

For general users, the system provides interpretable insights into cardiovascular health status and potential risk factors, enabling timely preventive actions and early intervention. For healthcare professionals, it functions as an effective remote monitoring tool, supporting continuous observation of ECG signals and facilitating the prompt identification of clinically significant abnormalities or temporal changes.

Moreover, owing to its low-cost implementation and edge-based architecture, the system can be readily deployed across diverse settings, including community healthcare facilities and resource-constrained environments, thereby improving the accessibility and scalability of cardiovascular disease screening and management.

3 Methodology and Implementation

3.1 System Overview

本作品硬體架構由 AD8232 ECG 感測模組、ESP32 與 Raspberry Pi 3B 組成。AD8232 負責單導程 ECG 訊號擷取，ESP32 負責資料取樣與傳輸，Raspberry Pi 3B 則負責訊號處理、特徵萃取、AF 偵測、風險推論與結果整合。此架構可在維持系統完整性的前提下兼顧成本、可攜性與邊緣部署需求。

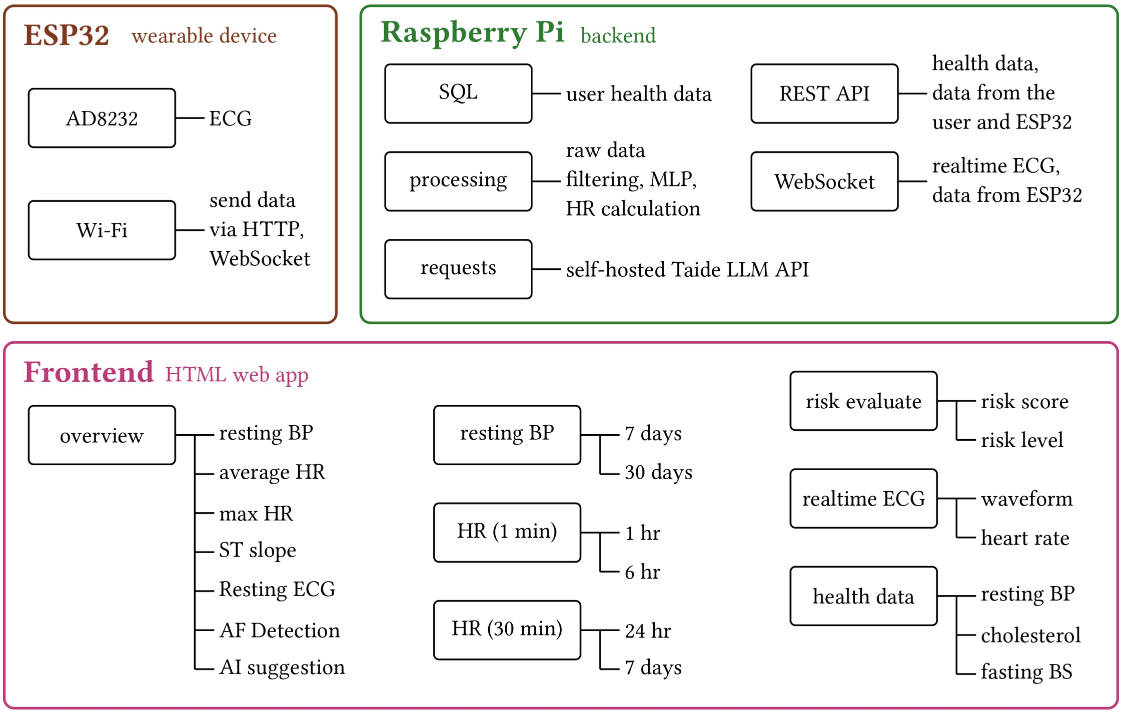

Figure 1: 作品模組全覽圖

而軟體流程的部分可分為四個主要模組：

1. 訊號處理：基於 Pan-Tompkins++ 演算法實作 R-peak 偵測，並加入 ST segment 相關分析 [2]。
2. 特徵萃取：由 R-peak 計算 MaxHR、Oldpeak、ST_Slope、RestingECG 等特徵，提供風險評估模型預測。
3. AF 偵測：以 RR interval 為基礎，使用 RdR map 與 non-empty cell 計數進行判斷。
4. 風險評估：以 CatBoost 為核心，根據使用者輸入資料是否完整，自動切換模型進行預測。

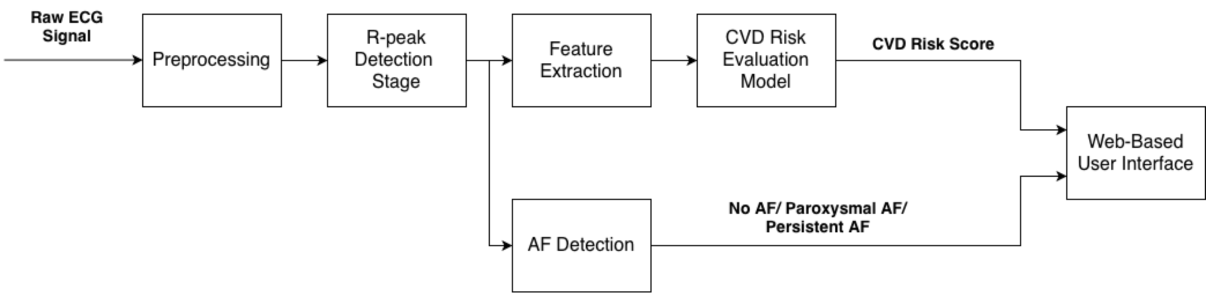

Figure 2: Data flow of the system

在模型部署方面，本作品保留兩組 CatBoost 模型。8-feature 模型作為部署版本，適用於僅有基本資料與 ECG 特徵的情況；10-feature 模型則於使用者同時提供 Cholesterol

與 FastingBS 等額外健康資料時啟用，以獲得較高辨識效能。兩組模型大小約 650KB 等級，可於邊緣裝置進行推論。

3.2 Cardiovascular Risk Prediction

3.2.1 Dataset and Problem Setup

We utilized cardiovascular disease datasets from the UCI repository, including the Cleveland, Hungary, Switzerland, and VA Long Beach databases, and further integrated them with the Stalog (Heart) dataset. After removing duplicate entries and records with missing values, the final dataset comprised 918 samples. It includes 14 input features and one target variable, as detailed in the following table.

Feature	Description
Age	Age of the patient (years)
Sex	Sex of the patient (M: Male, F: Female)
ChestPainType	Chest pain type (TA: Typical Angina, ATA: Atypical Angina, NAP: Non-Anginal Pain, ASY: Asymptomatic)
RestingBP	Resting blood pressure (mm Hg)
Cholesterol	Serum cholesterol (mm/dl)
FastingBS	Fasting blood sugar (1: >120 mg/dl, 0: otherwise)
RestingECG	Resting electrocardiogram results (Normal, ST: ST-T abnormality, LVH: left ventricular hypertrophy)
MaxHR	Maximum heart rate achieved (60–202)
ExerciseAngina	Exercise-induced angina (Y: Yes, N: No)
Oldpeak	ST depression value (numeric)
ST_Slope	Slope of peak exercise ST segment (Up, Flat, Down)
HeartDisease	Output class (1: Heart disease, 0: Normal)

Table 2: Summary of Input Features for Heart Disease Dataset

Among these 14 features, we finally selected 10 features for the deployed model since the remaining 4 features (Cholesterol, FastingBS, RestingBP, ExerciseAngina) are not directly measurable by our device and require user input. The 10 features used in the deployed

3.2.2 Preprocessing and Filters

Before extracting features, raw ECG signals must undergo noise filtering to ensure the accuracy of subsequent feature calculations. Our project initially utilized a 5-12 Hz filter to process the QRS segment. However, this frequency band over-suppressed the T-wave, causing the false removal of genuine ST depression or ST elevation features. To address this, we designed an independent 0.5-35 Hz filtering interval specifically for the ST Slope calculation, preserving complete and genuine ST segment features.

3.2.3 Feature Extraction Methods

Our processing pipeline primarily extracts four key features: Maximum Heart Rate, Oldpeak, ST Slope, and Resting Electrocardiogram (RestingECG). The R-peak measured by `Pan-Tompkins++` [2] serves as the reference point for calculating all subsequent features (like heart rate and segment windows). The extraction mechanisms, challenges, and optimization processes are detailed below.

3.2.3.1 Maximum Heart Rate

Definition: The maximum heart rate achieved during the measurement period, typically yielding a numeric value between 60 and 202.

Extraction Method: By utilizing the `Pan-Tompkins++` algorithm, we were able to detect R-peaks, which are subsequently converted into the heart rate.

3.2.3.2 Oldpeak

Definition: Oldpeak represents the ST depression caused by activity in comparison to rest. A value > 0 may indicate angina, while a value < 0 is potentially related to myocardial infarction. Oldpeak quantifies the absolute displacement of the ST segment relative to the electrically neutral PR baseline. In a healthy heart, the ST segment should be perfectly flush with the PR segment. When the heart muscle is stressed or deprived of oxygen (especially during the exercise phase of stress testing), the ST segment will deviate.

Extraction Method: It is calculated based on relative R-peak positions using the following formula:

$$\text{Oldpeak} = V_{\text{ST level}} - V_{\text{PR baseline}}$$

Here, *PR baseline* is the average potential between 200 ms and 120 ms prior to the R-peak, and *ST level* is the average potential between 100 ms and 160 ms after the R-peak. This timing window is crucial for accurately measuring the ST segment's displacement, and is suggested by multiple medical literature sources. [3], [4], [5]

3.2.3.3 ST Slope

Definition: The slope of the peak exercise ST segment, classified into three categories: Up, Flat, or Down. While the Oldpeak measures the absolute drop or rise of the ST segment, the

ST_Slope measures its morphological trajectory. How the ST segment behaves dynamically over time is a critical predictor of coronary artery disease.

Algorithmic Execution: Instead of taking a single average, the algorithm extracts all voltage data points within the defined ST Window and applies an Ordinary Least Squares linear regression to find the line of best fit:

$$V(t) = mt + b$$

Here, t represents time, V(t) is the voltage, b is the y-intercept, and m is the computed slope.

Diagnostic Classification: We categorize the ST slope into three categories: Up (upsloping), Flat, or Down (downsloping).

- *Upsloping*: $m > 0.5\frac{V}{s}$. (Often seen in normal, healthy exercise responses, even if slight ST depression is present).
- *Downsloping*: $m < -0.5\frac{V}{s}$. (A highly specific indicator of severe ischemic heart disease).
- *Flat*: $|m| \leq 0.5\frac{V}{s}$. (Also a strong indicator of ischemia, as a healthy ST segment should naturally slope upward into the T-wave).

3.2.3.4 Resting Electrocardiogram (RestingECG)

Definition: RestingECG is categorized into Normal, ST (having ST-T wave abnormality, such as T-wave inversion or ST elevation/depression > 0.05 mV), and LVH (showing probable or definite left ventricular hypertrophy by Estes' criteria).

T-wave Difficulties: T-waves are low-frequency, low-amplitude signals that are highly susceptible to baseline wandering, which is a type of low-frequency noise primarily caused by the user's respiration and minor electrode movements.

Solution:

1. Initial Attempt: To stabilize the T-wave for accurate inversion detection, we integrated a Wavelet Transform-based Baseline Wandering Removal algorithm (utilizing the open-source py-bwr library).
2. The Trade-off: While BWR successfully clarified the T-wave inversion morphology, testing revealed a critical flaw: the mathematical transformation aggressively warped the raw voltage levels, severely degrading the accuracy of ST elevation/depression measurements. Overall model accuracy dropped from 92.39% without BWR to 83.42% with global BWR.
3. Final Pipeline: To resolve this, the system employs a bifurcated processing strategy. Baseline correction is applied only in parallel streams specifically evaluating T-wave

morphology, while the original minimally filtered signal (0.5–35 Hz) is strictly preserved for calculating absolute ST segment offsets.

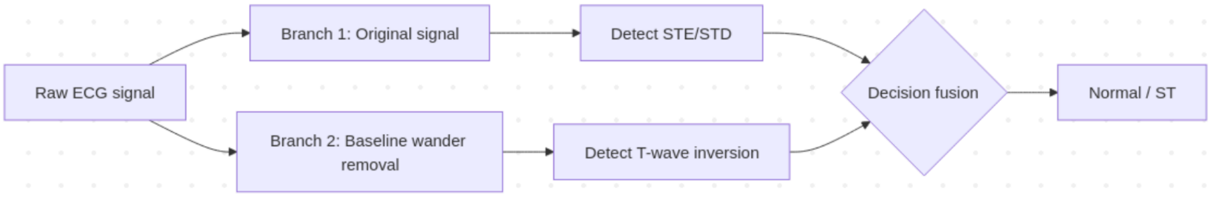

Figure 3: The bifurcated processing strategy for T-wave morphology correction.

LVH Difficulties: And for LVH, since it requires precordial leads—specifically V1, V5, and V6, we implemented a practical software-based workaround. During the initial profile registration on our web interface, the system proactively prompts users to input their known medical history, including any prior LVH diagnoses. This approach ensures that the machine learning model still receives the complete data array required for a comprehensive risk assessment.

3.2.3.5 Validation:

We validated the accuracy of the STE/STD feature by comparing it with the European ST-T Database. The accuracy of recognizing ST Elevation/Depression was found to be 99.16% and recall of ST Elevation/Depression was 91.6%.

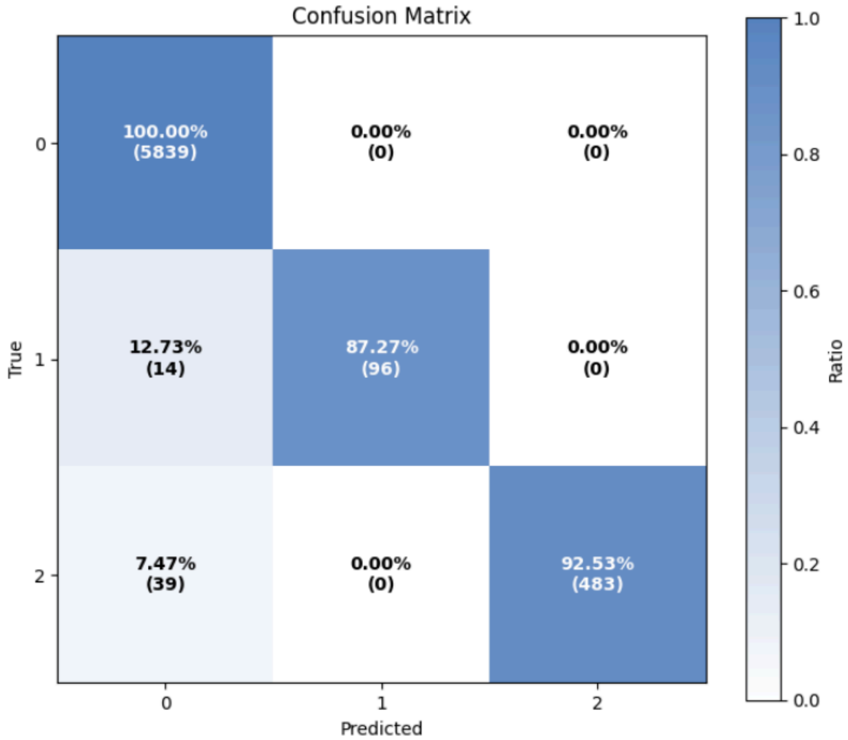

Figure 4: Confusion matrix for ST Elevation/Depression recognition.

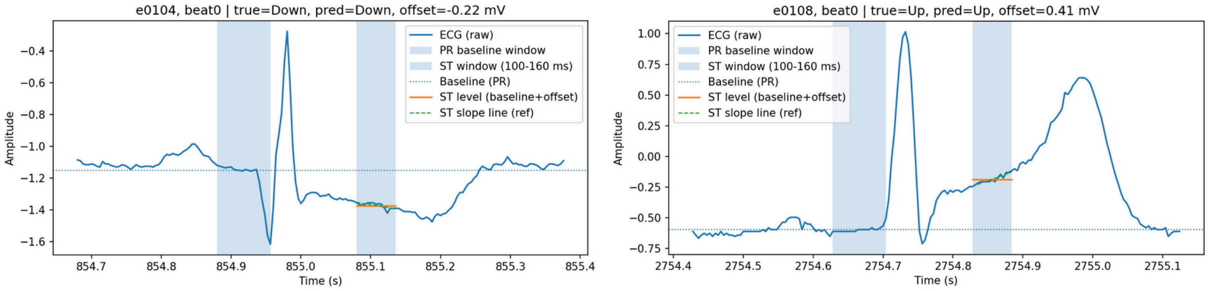

Figure 5: Examples of ECG signal with features annotated.

3.2.4 Model Training Methodology and Techniques

3.2.4.1 Feature Preprocessing

First, public heart disease datasets were reconstructed into a comprehensive labeled dataset (918 records) with standardized feature naming and data types. Categorical variables were transformed into numerical representations via label encoding, while missing numerical values were handled using mean imputation to prevent training bias resulting from data omissions. To address the issue of class imbalance, oversampling was employed during the training phase. This approach ensures a more balanced distribution of positive and negative classes, thereby enhancing the model's ability to identify minority classes.

3.2.4.2 Training

Regarding data partitioning, this study adopted an 80/20 holdout split (80% training set; 20% test set) consistent with existing literature, performing cross-validation and hyperparameter optimization within the training set. A progressive evolution strategy was applied to model development; initially, multiple baseline models were evaluated, including Logistic Regression, Gradient Boosting, CatBoost, and an average probability Ensemble model. Through comprehensive comparison, CatBoost exhibited the highest potential. Consequently, the final deployed model in this study evolved directly from the CatBoost baseline. By performing further fine-tuning, we effectively improved the consistency between offline evaluation and deployment behavior. The final deployed CatBoost model converged with the following hyperparameters: (iterations=600, learning_rate=0.1, depth=6, l2_leaf_reg=3, random_seed=42).

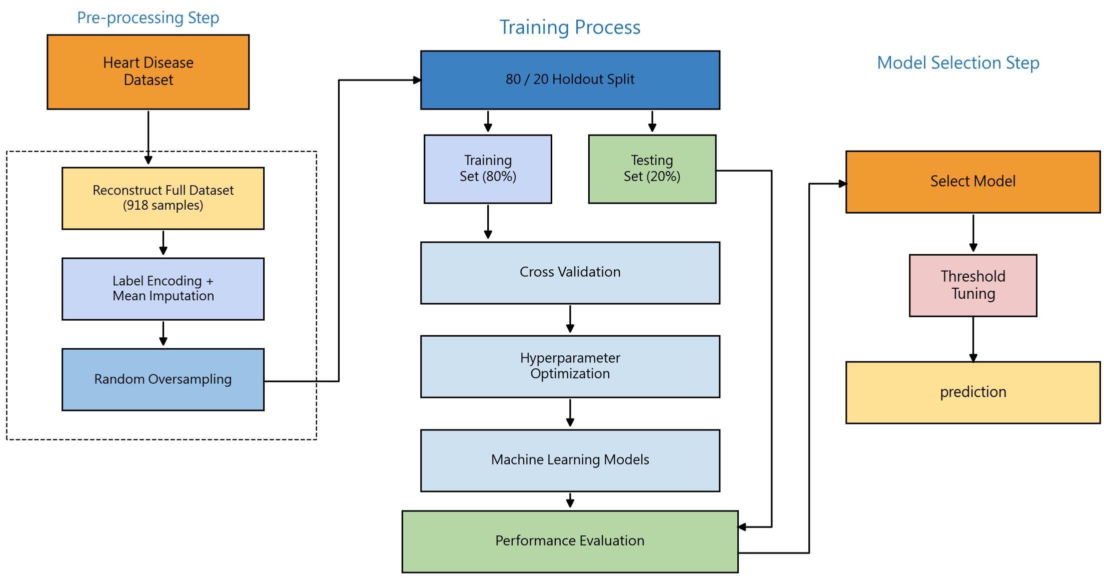

Figure 6: Flowchart of Model Development and Training

3.2.4.3 Classification

To balance prediction accuracy and system robustness, a Dual-model Routing mechanism was designed for the inference stage. When a user provides a complete set of clinical features, the system prioritizes the 10-Feature Model to deliver the best predictive performance. Conversely, if specific fields are missing, the system automatically switches to the 8-Feature Model, which utilizes data directly captured by our device's sensors to ensure stable and continuous output even under conditions of incomplete information.

3.2.4.4 Performance Evaluation

Accuracy, Precision, Recall, F1-score, and ROC-AUC were utilized as metrics for performance evaluation. Results on the holdout test set clearly illustrate the model's evolution trajectory: starting from the initial CatBoost baseline model (Accuracy 85.87%, Recall 85.29%), moving through fine-tuning, and culminating in threshold tuning. The final deployed 10-Feature model significantly pushed Accuracy and Recall to 96.57% and 97.06%, respectively.

Model	**Parameter**	**Accuracy**	**AUC**	**Precision**	**Recall**	**F1**
CatBoost	{verbose=False, random_state=369}	85.87	90.35	88.78	85.29	87.00
LogisticRegression	{max_iter=5000, random_state=369}	87.50	92.53	89.11	88.24	88.67
GradientBoosting	{random_state=369}	85.33	91.69	90.32	82.35	86.15
Ensemble		86.96	92.06	90.62	85.29	87.88
CatBoost after fine-tuning (before threshold tuning)	(iterations=600, learning_rate=0.1, depth=6, l2_leaf_reg=3, random_seed=42)	91.67	88.08	95.70	87.25	91.28

CatBoost after fine-tuning (after threshold tuning)	(iterations=600, learning_rate=0.1, depth=6, l2_leaf_reg=3, random_seed=42, threshold=0.20)	96.57	97.67	96.12	97.06	96.59

Table 3: Comparison of Model Performance Metrics on Test Set

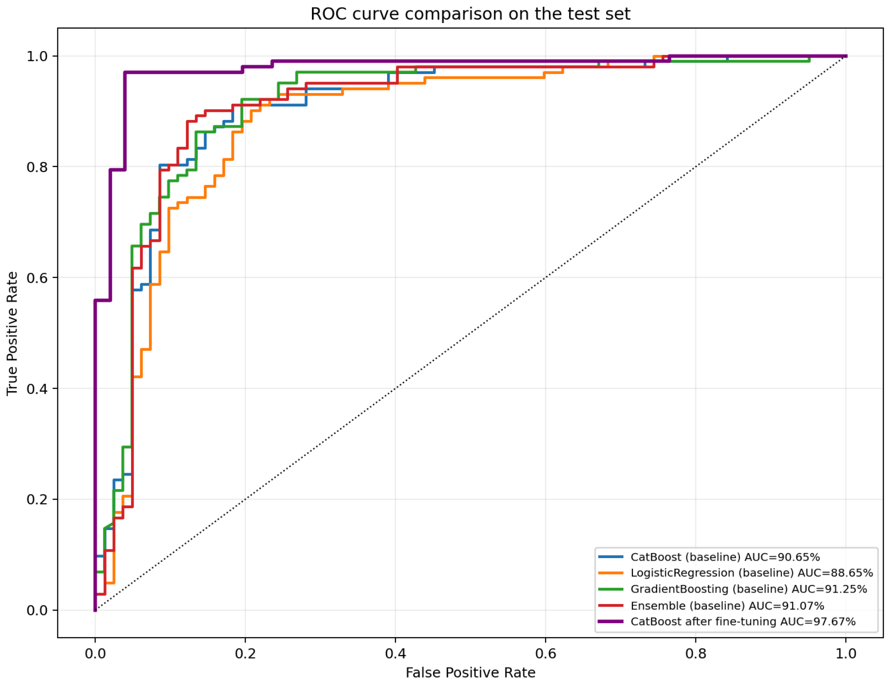

Figure 7: ROC curve comparison on test set

3.3 AF Detection Methodology and Validation

During the early development of the Atrial Fibrillation (AF) detection module, the Coefficient of Variation (CV) test method was initially employed [6]. While this method is fast to implement and computationally efficient, its decision criteria are fundamentally based on a single statistic. This makes it prone to misidentifying ectopic rhythms, such as Premature Ventricular Contractions (PVCs), as AF, leading to a high false alarm rate in practical applications.

To address this issue, a multi-feature integration approach proposed by Dash et al. [7] was subsequently explored, which utilizes TPR, RMSSD, and Shannon Entropy (SE) to improve discriminatory power. As shown in Table 2, this integrated approach achieved excellent classification results on the AFDB dataset. However, this method involves complex feature extraction and precise parameter tuning, incurring high engineering overhead and significantly longer computation times.

After a comprehensive evaluation of classification performance, false alarm control, algorithmic complexity, and maintenance costs, the RdR+NEC method [8] was ultimately selected for this work. This method offers the dual advantages of simple decision rules and extremely low computational complexity. As verified by the computation time comparison in Table 2, the current RdR+NEC version maintains high accuracy and balanced sensitivity and specificity while satisfying the requirements for real-time execution on edge devices. It has thus been adopted as the deployment method.

Table 2. Sensitivity, specificity and accuracy values for different methods from MIT-BIH dataset

Method	Sensitivity	Specificity	Accuracy	Computation time comparison
CV test	90.91	77.67	78.62	0.294 ms/window
RMSSD+TPR+SE	98.99	87.12	87.97	10.107 ms/window
RdR+NEC (current)	95.72	95.74	95.73	1.167 ms/window

The specific technical details and parameter settings of the RdR+NEC algorithm are as follows:

3.3.1 RdR Map Construction

Unlike the traditional Lorenz plot which utilizes a single dimension, this method maps successive RR intervals (RR_i) and their differences ($\mathrm{dRR}_i = \mathrm{RR}_i - \mathrm{RR}_{i-1}$) simultaneously onto a two-dimensional coordinate system. This design captures both heart rate and heart rate variability (HRV) information. During AF episodes, irregular rhythms cause data points to scatter randomly across a wide area of the map; conversely, normal or regular rhythms typically present as dense, localized clusters.

3.3.2 Gridding and Non-empty Cells (NEC)

The system divides the RdR map into a two-dimensional grid. For each input segment containing 128 cardiac cycles (128-beat window), the number of Non-empty Cells (NEC) is calculated. This count quantifies the degree of dispersion to classify the rhythm as AF or non-AF.

3.3.3 Threshold Determination

Based on the optimization analysis in [8], an NEC threshold of 65 was determined to provide optimal performance for a 128-beat window configuration.

3.3.4 Edge Deployment Advantages

Compared to methods requiring extensive feature extraction or complex floating-point operations, RdR+NEC relies solely on a single integer count of NECs for classification. Its low computational overhead enables high-precision monitoring while satisfying the real-time and long-term monitoring requirements of battery-powered wearable devices.

Figure 8: Schematic Diagram of Atrial Fibrillation (AF) Detection

To confirm the algorithm's generalization capabilities, verification was conducted across multiple public datasets. As indicated in Table 3, the RdR+NEC method demonstrates highly stable and superior detection performance across diverse data distributions.

Table 3. Performance validation results of the RdR+NEC method across public datasets

Database	**Sensitivity**	**Specificity**
NSRDB	NA*	88.10
NSR2DB	NA*	96.30
AFDB	95.70	95.70
Combined database	95.70	94.20

* NA: Sensitivity is not applicable because no true AF episodes are present.

3.4 TAIDE-based health assistant

To enhance user experience and provide personalized health insights, we integrated an advanced AI assistant directly into the frontend interface.

For the core language model, we selected **Gemma-3-TAIDE-12b-Chat**, a robust 12-billion parameter model optimized by the TAIDE project. This specific model was chosen for its excellent reasoning capabilities and its strong proficiency in handling Traditional Chinese, ensuring that the health advice and explanations provided to users are both accurate and natural to read.

Due to the substantial computational requirements of running a 12B parameter model, we deployed the language model as a dedicated server hosted on a Spark DGX system. This backend server handles all the heavy AI inference tasks. When a user requests health insights on the frontend application, the system makes a call to the server, retrieving the AI-generated responses efficiently. This client-server architecture ensures that the frontend remains highly responsive and accessible, without being burdened by the intensive processing demands of the large language model.

Figure 9: TAIDE-based health assistant response

3.5 System Deployment, Cost Analysis, and Feasibility

To evaluate the practical feasibility of the proposed system, we analyze the hardware cost structure and deployment requirements.

The system consists of the following key components:

Component	Function	Cost (TWD)
AD8232 ECG module	ECG signal acquisition	200
ESP32	Data acquisition and transmission	150
Raspberry Pi 3B	Edge processing	1500
Power module & peripherals	Battery and supporting components	100

Total		< 2000

Table 4: Hardware Cost Breakdown of the Proposed System

The total system cost is maintained below 2000 TWD, which is significantly lower than most commercial wearable ECG devices. From a system perspective, the proposed design achieves a favorable trade-off between computational capability, model complexity, and deployment cost. The use of lightweight machine learning models (approximately 650 KB) enables real-time inference on edge devices without requiring specialized hardware acceleration. Compared to smartwatch-based ECG systems, which typically rely on proprietary hardware ecosystems and external processing pipelines, the proposed system provides a transparent and cost-efficient architecture with full control over signal processing and inference. These results indicate that the proposed system is both technically effective and economically viable for large-scale deployment in home healthcare and remote monitoring scenarios.

3.6 Prototype Implementation and System Demonstration

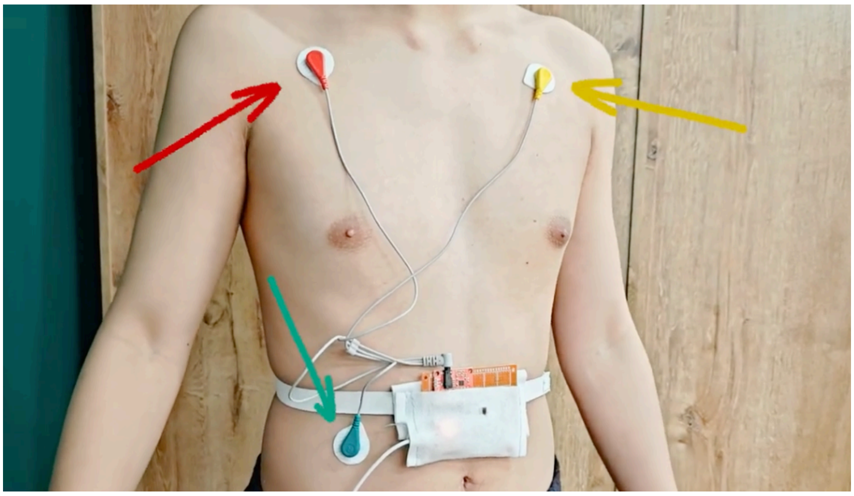

Figure 10: Wearable ECG System Configuration and Electrode Placement

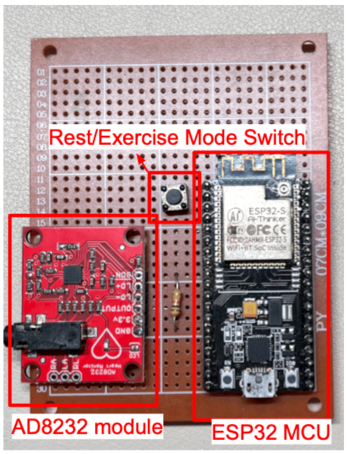

Figure 11: Hardware Implementation of the Wearable ECG Monitoring System

Figure 12: Frontend interface

4 技術限制、風險與可信度說明

為提高作品可信度，本作品亦主動說明現階段限制如下：

1. 目前主要使用單導程 ECG，與醫療級多導程系統相比，資訊量仍有限。
2. 心臟疾病風險評估模型主要基於公開資料集驗證，未來仍需更多真實使用場景驗證。
3. 心房顫動 (AF) 偵測目前以公開資料庫與原型資料進行驗證，未來仍可補充更大規模使用者測試。
4. LLM 健康建議若作為後續功能，僅應視為輔助性資訊，不應取代醫師診斷。
5. 本系統定位為健康管理與早期警示工具，而非醫療診斷設備。

5 Conclusion and Future Work

The system can be further extended to support multi-lead ECG acquisition, thereby enriching the information available for cardiovascular risk assessment. In addition, the current wired configuration and reliance on gel-based electrodes may limit user comfort and usability. Future work may therefore investigate wireless electrode designs or dry electrode technologies to enhance wearability and support long-term use. Furthermore, the system could be integrated with cloud-based services to enable home-based diagnosis and remote monitoring, allowing users to share ECG data and risk assessment results with healthcare professionals for more comprehensive and continuous care.

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